

Лекция 7

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$$df = \underbrace{\left(dx_1 \frac{\partial}{\partial x_1} + \dots + dx^n \frac{\partial}{\partial x^n} \right)}_{\hat{A}} f$$
$$\hat{A}(\hat{A}f) = \hat{A}(df) = d(df) = d^2f$$

$$d^k f = \left(dx^1 \frac{\partial}{\partial x^1} + \dots + dx^n \frac{\partial}{\partial x^n} \right)^k f$$
$$\hat{A}(d^k f) = d(d^k f) = d^{k+1} f$$

Зам. $f: U \subset \mathbb{R}^n \rightarrow \mathbb{R}$
 f имеет в $m. (x_0, y_0)$

непр. частные производ. k -го порядка

$$d^k f(x_0) = \sum_{m=0}^k \frac{\partial^k f(x_0, y_0)}{\partial^m x \partial^{k-m} y} d^m x d^{k-m} y \quad \bullet C_k^m$$

Дока-во: $d^k f = \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right)^k f$ ---

$$\left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right) \left(dx \frac{\partial}{\partial x} + dy \frac{\partial}{\partial y} \right)$$

$$d^k f(x_0, y_0) = \left(\sum_{m=0}^k dx^m \frac{\partial^m}{\partial x^m} \cdot dy^{k-m} \frac{\partial^{k-m}}{\partial y^{k-m}} \right) f \Big|_{(x,y)=(x_0,y_0)} \quad \ominus$$

$$\ominus \sum_{k=0}^k C_k^m \frac{\partial^k f(x_0, y_0)}{\partial^m x \partial^{k-m} y} \cdot d^m x d^{k-m} y$$

Зам. $f: U \subset \mathbb{R}^n \rightarrow \mathbb{R}$
 f $(m+1)$ -раз глф. в $U \subset \mathbb{R}^n \Rightarrow$

$$\forall \bar{x} \in U \subset \mathbb{R}^n \exists \Theta \in (0,1):$$
$$f(x) = \sum_{k=0}^n \frac{d^k f(x_0)}{k!} + \frac{d^{n+1} f(\bar{x}_0 + \Theta \Delta \bar{x})}{(n+1)!}; \Delta \bar{x} = \bar{x} - \bar{x}_0$$

Дока-во: $\varphi(t) = f(\bar{x}_0 + t \Delta \bar{x})$

$f: \mathbb{R}^n \rightarrow \mathbb{R}^1$
 $g: \mathbb{R}^1 \rightarrow \mathbb{R}^n$
 $h(x) = g(f(x))$
 $\partial h = \partial g \partial f$

$g(t) = \bar{x}_0 + t \Delta \bar{x}$
 $\partial g = (\Delta x^1, \dots, \Delta x^n)^T$
 $\partial h = \left(\frac{\partial f}{\partial x^1} \Delta x^1, \dots, \frac{\partial f}{\partial x^n} \Delta x^n \right)$



$$\varphi'(t) = dx^1 \frac{\partial f}{\partial x^1} (\bar{x}_0 + t \Delta \bar{x}) + \dots + dx^n \frac{\partial f}{\partial x^n} (\bar{x}_0 + t \Delta \bar{x})$$

$$\hat{A} = dx^1 \frac{\partial}{\partial x^1} + \dots + dx^n \frac{\partial}{\partial x^n} \quad \varphi'(t) = (\hat{A} f)(\bar{x}_0 + t \Delta \bar{x})$$

$$\varphi(t) = \varphi(0) + \sum_{k=1}^n \frac{\varphi^{(k)}(0)}{k!} t^k + \frac{\varphi^{(n+1)}(\bar{x}_0 + \Theta \Delta \bar{x})}{(n+1)!}$$

$$f(\bar{x}) = f(\bar{x}_0) + \sum_{k=1}^n \frac{d^k f(\bar{x}_0)}{k!} + \frac{d^{n+1} f(\bar{x}_0 + \Theta \Delta \bar{x})}{(n+1)!}$$

Зам. $f: \mathbb{R}^n \supset U \subset \mathbb{R}^n \rightarrow \mathbb{R}$

Все частные производ. n -го порядка
непр. в $m. \bar{x}_0$
$$f(x) = \sum_{k=0}^n \frac{d^k f(\bar{x}_0)}{k!} + o(|\Delta x|^n) \quad \text{при } \Delta x \rightarrow 0$$

Дока-во: f глф-на n раз в $U \subset \mathbb{R}^n$

по φ -ле Тейлора $\subset$ осс. n -мем в форме
Лагранжа: $f(\bar{x}) = \sum_{k=0}^n \frac{d^k f(x_0)}{k!} + \frac{d^{n+1} f(\bar{x}_0 + \Theta \Delta \bar{x})}{(n+1)!} \quad \Theta \in (0,1)$

$$f(\bar{x}) = \sum_{k=0}^n \frac{d^k f(\bar{x}_0)}{k!} + \frac{d^{n+1} f(\bar{x}_0 + \Theta \Delta \bar{x})}{(n+1)!} - \frac{d^n f(\bar{x}_0)}{n!}$$
$$|d^n f(\bar{x}_0 + \Theta \Delta \bar{x}) - d^n f(\bar{x}_0)| = \left| \sum_{i_1=1}^n \dots \sum_{i_n=1}^n \frac{\partial^n f(\bar{x}_0 + \Theta \Delta \bar{x})}{\partial x^{i_1} \dots \partial x^{i_n}} dx^{i_1} \dots dx^{i_n} - \sum_{i_1=1}^n \dots \sum_{i_n=1}^n \frac{\partial^n f(\bar{x}_0)}{\partial x^{i_1} \dots \partial x^{i_n}} dx^{i_1} \dots dx^{i_n} \right|$$
$$\underbrace{\frac{|\Delta x^{i_k}|}{\Delta x} \leq 1}_{=1} \left| \frac{d^n f(\bar{x}_0 + \Theta \Delta \bar{x}) - d^n f(\bar{x}_0)}{|\Delta \bar{x}|^n} \right| \leq \left| \sum_{i_1=1}^n \dots \sum_{i_n=1}^n \left(\frac{\partial^n f(\bar{x}_0 + \Theta \Delta \bar{x})}{\partial x^{i_1} \dots \partial x^{i_n}} - \frac{\partial^n f(\bar{x}_0)}{\partial x^{i_1} \dots \partial x^{i_n}} \right) \right| \xrightarrow{|\Delta \bar{x}| \rightarrow 0} 0$$

σ -алгебра σ -алгебра σ -алгебра

Опр. $\mathcal{P} = \{ \bar{x} \in \mathbb{R}^k : a^k \leq x^k \leq b^k \}$, где $a^k \leq b^k$
кажд. клетка (длина) $a^k, b^k \in \mathbb{R}^1$
 $k \in \{1, \dots, n\}$

$$\mu(\mathcal{P}) = \prod_{k=1}^n (b^k - a^k)$$
 — мера клетки $\mathcal{P}$

Опр. $A \subset \mathbb{R}^n$ — клеточное мн-во / σ -алгебра мн-во /
элементарные мн-ва, если $\exists n_1, \dots, n_k$ клеток
 $int \mathcal{P}_k \cap int \mathcal{P}_j = \emptyset \quad \forall k \neq j$
и $A = \bigcup_{k=1}^m \mathcal{P}_k$
$$\mu(A) = \sum_{k=1}^m \mu(\mathcal{P}_k)$$

Зам. мера элементарного множества определена корректно.

Зам. A, B клеточные, $\exists n: A \cap B, A \cup B$
 $A \setminus int B$ — клеточные

уст. A, B клеточные
 $int A \cap int B = \emptyset \Rightarrow \mu(A \cup B) = \mu(A) + \mu(B)$

если $A \cap B \neq \emptyset$, то $\mu(A \cup B) \leq \mu(A) + \mu(B)$

Опр. $A \subset \mathbb{R}^n$
$$\mu^*(A) = \inf_{\substack{X \supset A \\ X \text{ элементар.}}} \mu(X)$$
 (внешняя мера Жордана)
$$\mu_*(A) = \sup_{\substack{X \subset A \\ X \text{ элементар.}}} \mu(X)$$
 (внутренняя мера Жордана)

Если $\mu_*(A) = \mu^*(A)$ A измерим по Жордану
$$\mu(A) = \mu_*(A) = \mu^*(A)$$
 мера Жордана множества A

$$A = [0, 1] \cap \mathbb{Q} \quad \partial A = [0, 1] \quad \mu(A) = 1$$

$$\mu^*(A) \geq 1$$

$$\mu_*(A) = 0$$